

```
In [1]: # Import Required Libraries

import arcpy
import pandas as pd
import matplotlib.pyplot as plt
from arcpy.sa import *

print(arcpy.__version__)
```

3.5

```
In [2]: # Enable Spatial Analyst

arcpy.CheckOutExtension("Spatial")
```

Out[2]: 'CheckedOut'

```
In [3]: # Define Your Inputs
# For each updated automation of this notebook, change "Landuse_raster" and "bounda

year = 2024

landuse_raster = r"C:\Users\User\Documents\Bootcamp GIS Certification\5_Setup AWS m
boundary_fc = r"C:\Users\User\Documents\Bootcamp GIS Certification\5_Setup AWS map
output_folder = r"C:\Users\User\Documents\Bootcamp GIS Certification\5_Setup AWS ma
```

```
In [4]: # Set Workspace

arcpy.env.workspace = output_folder
arcpy.env.overwriteOutput = True
```

```
In [5]: # Clip the Land Use Raster

clipped_raster = f"{output_folder}/landuse_clip_{year}.tif"

arcpy.Clip_management(
    landuse_raster,
    "#",
    clipped_raster,
    boundary_fc,
    "#",
    "ClippingGeometry"
)
```

Out[5]:

Messages

Start Time: Monday, March 9, 2026 9:28:35 AM

Building Pyramids...

Succeeded at Monday, March 9, 2026 9:28:39 AM (Elapsed Time: 3.85 seconds)

```
In [6]: # Find a field in the attribute table that has data for every data point (like OBJECTID)

fields = [f.name for f in arcpy.ListFields(boundary_fc)]
print(fields)
```

```
['FID', 'ADMINFORES', 'REGION', 'FORESTNUMB', 'FORESTORGC', 'FORESTNAME', 'GIS_ACRE_S', 'SHAPE_AREA', 'SHAPE_LEN', 'Shape__Area', 'Shape__Length', 'Shape']
```

```
In [7]: # Calculate Area per Land Use using the FID field

table_output = f"{output_folder}/landuse_area_{year}.dbf"

TabulateArea(
    boundary_fc,
    "FID",
    clipped_raster,
    "Value",
    table_output
)
```

Out[7]:

Messages

Start Time: Monday, March 9, 2026 9:28:39 AM

Input class raster being used as snap raster; this is new default behavior.

Succeeded at Monday, March 9, 2026 9:28:41 AM (Elapsed Time: 1.95 seconds)

```
In [8]: # Convert Table to a Pandas DataFrame

arr = arcpy.da.TableToNumPyArray(table_output, "")
df = pd.DataFrame(arr)

df
```

```
Out[8]:
```

	OID	FID_	VALUE_11	VALUE_21	VALUE_22	VALUE_23	VALUE_24	VALUE_31	VALU
0	0	1	2449800.0	43425900.0	13314600.0	3184200.0	865800.0	2064600.0	303134



```
In [9]: # Convert pixels to acres

pixel_area_m2 = 30 * 30
acre_conversion = 4046.86

for col in df.columns:
    if "VALUE_" in col:
        df[col] = (df[col] * pixel_area_m2) / acre_conversion

df
```

Out[9]:

	OID	FID_	VALUE_11	VALUE_21	VALUE_22	VALUE_23	VALUE_24	
0	0	1	544822.405519	9.657688e+06	2.961096e+06	708149.034066	192549.285125	4



```
In [10]: # Calculate Percentages

total_acres = df.filter(like="VALUE_").sum(axis=1)

percent_df = df.filter(like="VALUE_").div(total_acres, axis=0) * 100

percent_df
```

Out[10]:

	VALUE_11	VALUE_21	VALUE_22	VALUE_23	VALUE_24	VALUE_31	VALUE_41	VALUE_42
0	0.27552	4.883952	1.497444	0.358115	0.097373	0.232198	34.092414	29.374968



```
In [11]: # Create a Summary Table

summary = pd.DataFrame({
    "LandUse": df.filter(like="VALUE_").columns,
    "Acres": df.filter(like="VALUE_").iloc[0].values,
    "Percent": percent_df.iloc[0].values
})

summary
```

Out[11]:

	LandUse	Acres	Percent
0	VALUE_11	5.448224e+05	0.275520
1	VALUE_21	9.657688e+06	4.883952
2	VALUE_22	2.961096e+06	1.497444
3	VALUE_23	7.081490e+05	0.358115
4	VALUE_24	1.925493e+05	0.097373
5	VALUE_31	4.591560e+05	0.232198
6	VALUE_41	6.741547e+07	34.092414
7	VALUE_42	5.808704e+07	29.374968
8	VALUE_43	2.572154e+07	13.007541
9	VALUE_52	8.595464e+06	4.346779
10	VALUE_71	9.918690e+06	5.015942
11	VALUE_81	1.112122e+07	5.624070
12	VALUE_82	1.255774e+06	0.635052
13	VALUE_90	1.087043e+06	0.549724
14	VALUE_95	1.761366e+04	0.008907

In [12]: *# Create readable land use labels*
Combine multiple fields into one

```
landuse_labels = {
    "VALUE_11": "Open Water",
    "VALUE_21": "Developed",
    "VALUE_22": "Developed",
    "VALUE_23": "Developed",
    "VALUE_24": "Developed",
    "VALUE_31": "Other",
    "VALUE_41": "Deciduous Forest",
    "VALUE_42": "Evergreen Forest",
    "VALUE_43": "Deciduous Forest",
    "VALUE_52": "Grassland/Scrub Shrub",
    "VALUE_71": "Grassland/Scrub Shrub",
    "VALUE_81": "Farmland",
    "VALUE_82": "Farmland",
    "VALUE_90": "Wetlands",
    "VALUE_95": "Wetlands"
}

summary = pd.DataFrame({
    "LandUse": df.filter(like="VALUE_").columns,
    "Acres": df.filter(like="VALUE_").iloc[0].values,
    "Percent": percent_df.iloc[0].values
})
```

```

summary["LandUse"] = summary["LandUse"].map(landuse_labels)

# Combine developed classes
summary = summary.groupby("LandUse", as_index=False).sum()

# Reduce decimal places / round numbers for easier readability
summary["Acres"] = summary["Acres"].round(0)
summary["Percent"] = summary["Percent"].round(2)

# SORT largest to smallest
summary = summary.sort_values(by="Acres", ascending=False)

summary

```

Out[12]:

	LandUse	Acres	Percent
0	Deciduous Forest	93137010.0	47.10
2	Evergreen Forest	58087035.0	29.37
4	Grassland/Scrub Shrub	18514154.0	9.36
1	Developed	13519482.0	6.84
3	Farmland	12376996.0	6.26
7	Wetlands	1104656.0	0.56
5	Open Water	544822.0	0.28
6	Other	459156.0	0.23

In [13]: # Create the Chart

```

plt.figure()

plt.bar(summary["LandUse"], summary["Percent"])

plt.xticks(rotation=45, ha="right")

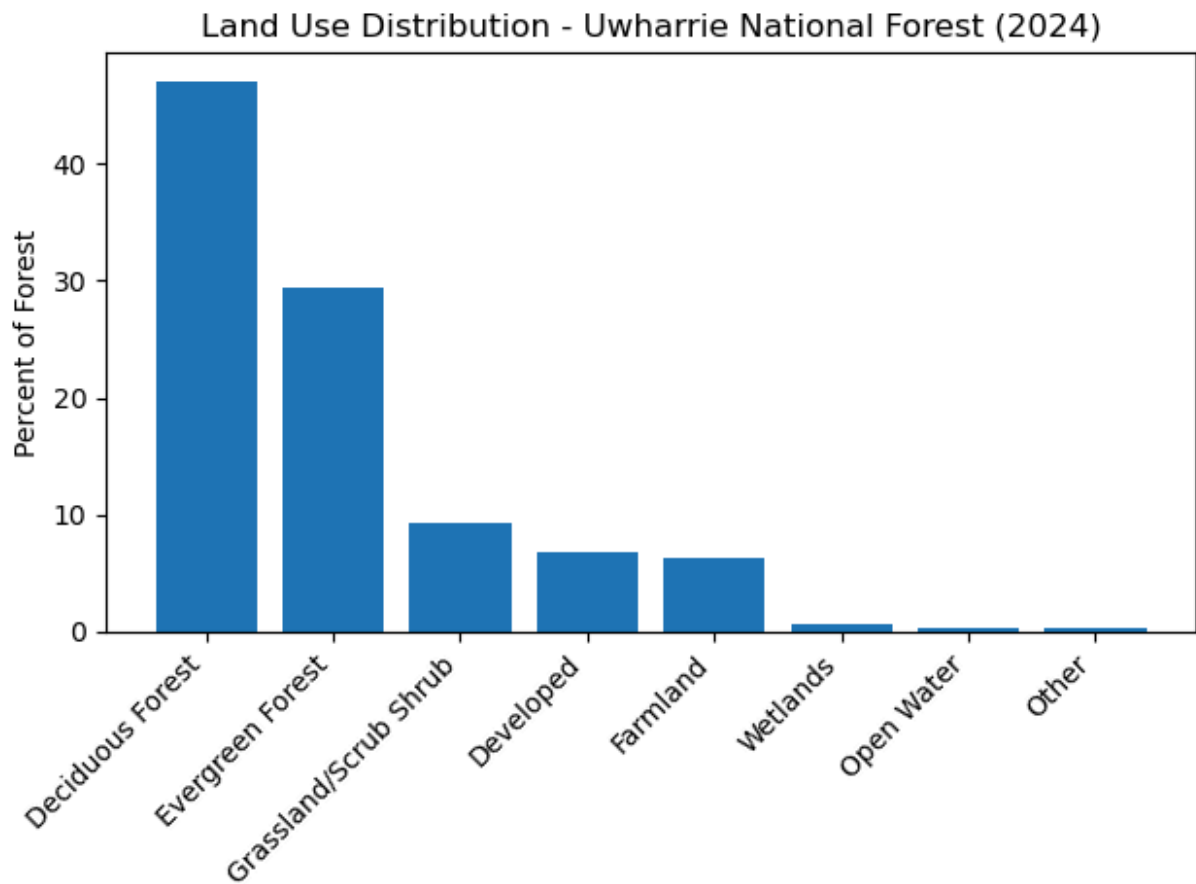
plt.ylabel("Percent of Forest")

plt.title(f"Land Use Distribution - Uwharrie National Forest ({year})")

plt.tight_layout()

plt.show()

```



```
In [14]: # Save the Table
summary.to_csv(f"{output_folder}/landuse_summary_{year}.csv", index=False)

#Save the Chart
plt.figure()

plt.bar(summary["LandUse"], summary["Percent"])

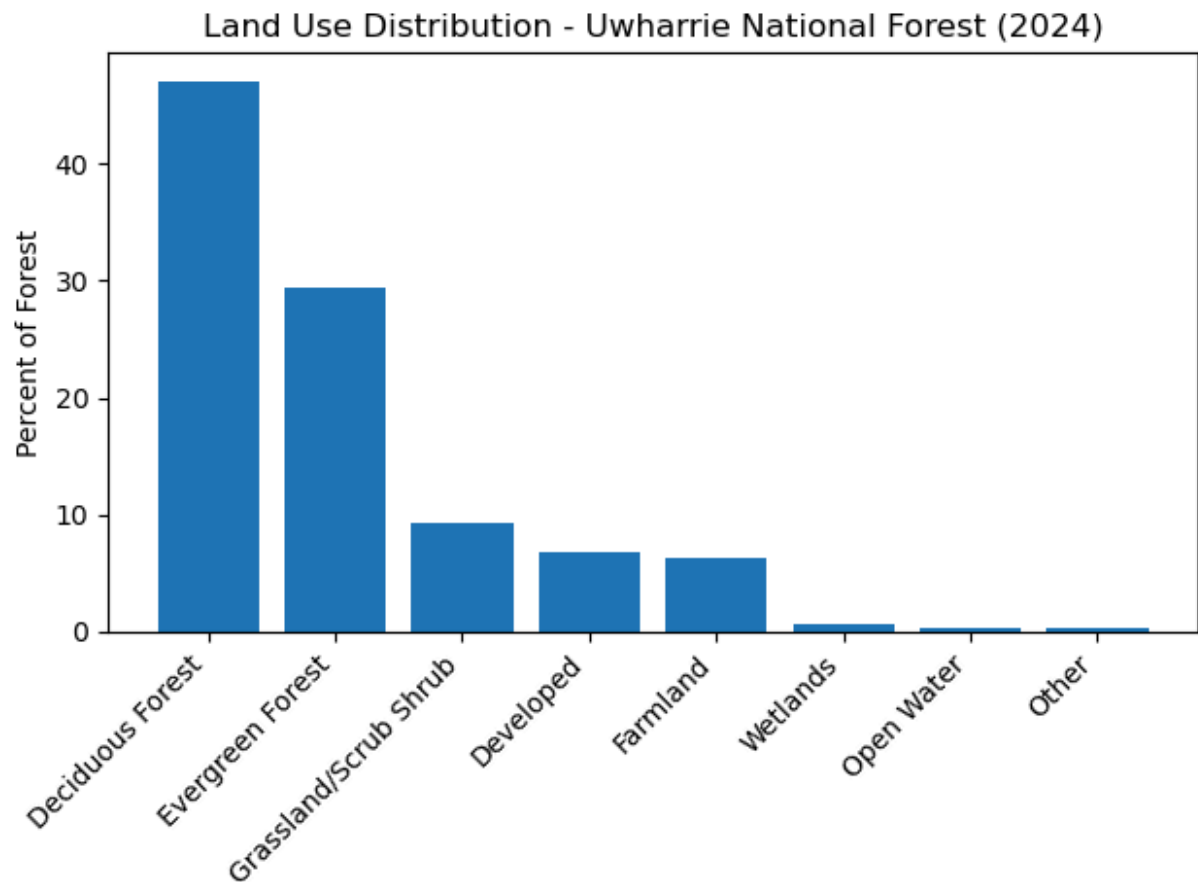
plt.xticks(rotation=45, ha="right")

plt.ylabel("Percent of Forest")

plt.title(f"Land Use Distribution - Uwharrie National Forest ({year})")

plt.tight_layout()

plt.savefig(f"{output_folder}/landuse_chart_{year}.png")
```



```
In [15]: # Export the Notebook to PDF

import subprocess

subprocess.run([
    "jupyter",
    "nbconvert",
    "--to",
    "pdf",
    "Uwharrie_LandUse_Analysis.ipynb"
])
```

```
Out[15]: CompletedProcess(args=['jupyter', 'nbconvert', '--to', 'pdf', 'Uwharrie_LandUse_Analysis.ipynb'], returncode=1)
```

```
In [16]: # Verify files exist

import os

os.listdir()
```

```
Out[16]: ['.anaconda',
'.conda',
'.condarc',
'.continuum',
'.dotnet',
'.ipynb_checkpoints',
'.ipython',
'.jupyter',
'.matplotlib',
'.virtual_documents',
'anaconda3',
'anaconda_projects',
'AppData',
'Application Data',
'Contacts',
'Cookies',
'Desktop',
'Documents',
'Downloads',
'Favorites',
'GIS',
'Links',
'Local Settings',
'Music',
'My Documents',
'NetHood',
'Notebooks',
'NTUSER.DAT',
'ntuser.dat.LOG1',
'ntuser.dat.LOG2',
'NTUSER.DAT{17e1b4b6-011c-11f1-aa1a-b03d76f40af6}.TM.blf',
'NTUSER.DAT{17e1b4b6-011c-11f1-aa1a-b03d76f40af6}.TMContainer00000000000000000000',
1.regtrans-ms',
'NTUSER.DAT{17e1b4b6-011c-11f1-aa1a-b03d76f40af6}.TMContainer00000000000000000000',
2.regtrans-ms',
'ntuser.ini',
'OneDrive',
'Pictures',
'PrintHood',
'Recent',
'Saved Games',
'Searches',
'SendTo',
'Start Menu',
'Templates',
'Untitled.ipynb',
'Uwharrie_LandUse_Analysis.ipynb',
'Videos']
```

```
In [1]: import subprocess
subprocess.run(["git", "--version"])
```

```
Out[1]: CompletedProcess(args=['git', '--version'], returncode=0)
```

In [2]: *#Initialize Git*

```
import subprocess
subprocess.run(["git", "init"])
```

Out[2]: CompletedProcess(args=['git', 'init'], returncode=0)

In [3]: *# Configure your Git Identity*

```
subprocess.run(["git", "config", "--global", "ijones17", "Isaak Jones"])
subprocess.run(["git", "config", "--global", "isaakjones2@gmail.com", "isaakjones2@
```

Out[3]: CompletedProcess(args=['git', 'config', '--global', 'isaakjones2@gmail.com', 'isaakjones2@gmail.com'], returncode=1)

In [4]: *# Connect to your Repository*

```
subprocess.run([
    "git",
    "remote",
    "add",
    "origin",
    "https://github.com/ijones17/GISbootcamp.git"
])
```

Out[4]: CompletedProcess(args=['git', 'remote', 'add', 'origin', 'https://github.com/ijones17/GISbootcamp.git'], returncode=0)

In [5]: *# Add your PDF File that you created of this notebook*

```
subprocess.run(["git", "add", "Uwharrie_LandUse_Analysis.ipynb"])
subprocess.run(["git", "add", "Uwharrie_LandUse_Analysis.pdf"])
```

Out[5]: CompletedProcess(args=['git', 'add', 'Uwharrie_LandUse_Analysis.pdf'], returncode=128)

In [6]: *# Commit the files*

```
subprocess.run([
    "git",
    "commit",
    "-m",
    "Automated Uwharrie National Forest land use analysis notebook"
])
```

Out[6]: CompletedProcess(args=['git', 'commit', '-m', 'Automated Uwharrie National Forest land use analysis notebook'], returncode=128)

In [7]: *# Push to GitHub*

```
subprocess.run(["git", "branch", "-M", "main"])

subprocess.run([
    "git",
    "push",
```

```
"-u",  
"origin",  
"main"  
])
```

Out[7]: CompletedProcess(args=['git', 'push', '-u', 'origin', 'main'], returncode=1)

In []: